Process Characterization Plan

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCP-007

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Field	Value
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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-02-02	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
AMV	analytical method validation
AU	absorbance unit
BLA	Biologics License Application
Cpk	process capability index
CQA	critical quality attribute
DNA	deoxyribonucleic acid
DoE	design of experiments
DS	drug substance
ELISA	enzyme-linked immunosorbent assay
HCP	host cell protein
HMW	high molecular weight species
HPLC	high performance liquid chromatography
IAC	intermediate acceptance criterion
IgG1	immunoglobulin G, subclass 1
LMW	low molecular weight species
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
OD280	optical density at 280 nm
PAR	proven acceptable range
PRESS	prediction error sum of squares
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
SEC	size exclusion chromatography
SOP	standard operating procedure

1 Purpose and scope

The A-Mab drug substance process is being transferred from the development site in Cambridge, MA, to the commercial drug substance site in Grafton, WI, under the Process Transfer Plan PTP-001. This plan describes the Stage 1 characterization of the cation exchange chromatography step, which is Step 7 of the drug substance train and its principal polishing operation. The studies will be executed on a scale-down model of the commercial column, in support of a process that operates at a 15,000 L production bioreactor scale at the receiving site. Their purpose is to measure how far the aggregate content, the host cell protein content, the fragment content and the yield of the eluate pool move as each parameter is varied across the ranges in Table 6.1, to define the region over which the pool meets its criteria, and to justify the ranges and the controls that will be applied in commercial manufacture (U.S. Food and Drug Administration 2011).

The scope of this plan covers the 5 process parameters of the step, the size variant and impurity attributes the step clears, and the step yield. Four parameters will be studied in a multivariate design and one will be assessed one at a time, on the assignment made in the risk assessment RA-001. The feed to the step is the neutralized pool of the low-pH viral inactivation step, and its attributes are treated as inputs to the study rather than as factors of it.

Three activities lie outside this plan. Resin life cycle, sanitization and repacking are qualified separately under SOP-2008, and the studies here will be run on resin inside that qualified life. Validation of the analytical methods is covered by the method validation reports in Table 3.2, and this plan uses those methods as validated. Confirmation of the ranges at commercial scale belongs to Stage 2 and to the process performance qualification batches, which follow the transfer.

No viral clearance claim is made for the cation exchange step. Clearance of enveloped virus is claimed at the low-pH viral inactivation step, the anion exchange step and the virus filtration step, and clearance of small viruses at the anion exchange step and the virus filtration step. The cumulative claim is consolidated in the master report PCMR-001 (International Council for Harmonisation 2023a). Cation exchange contributes clearance of host cell protein, DNA and leached Protein A, and those contributions are within scope.

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2026-02-02, before the first study of the step started on 2026-02-04. The studies it governs ran until 2026-03-27 (Section 11). Version 2.0, effective 2026-09-30, is issued after them under the Quality Assurance (QA)

investigation INV-001 (Acceptance criteria for proven-acceptable-range analysis set after data review). QA opened the investigation on 2026-08-03, after reviewing the master report PCMR-001 Version 1.0 for the Biologics License Application (BLA). It closed the investigation on 2026-09-18 with the classification major. INV-001 found that the criteria used to judge each step in the proven acceptable range (PAR) analysis had been set after the data were seen and outside the approved protocols. This revision carries out its corrective action CA2, the amendment of the step plans with the criteria and the fragment method.

The plan is reissued under INV-002 as well (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0). That investigation examined the response-surface models of the campaign that fall below the model adequacy criteria of the master plan PCMP-001 §7.2. Its corrective actions are the augmentation of the designs that its rule names (CA1), the refit of every other failing model (CA2) and the selection of the model of every output (CA3). The reports apply them, PCR-007 for this step, and this plan does not.

This revision keeps the design and the sampling of Version 1.0 of this plan, and the wording of every Version 1.0 decision rule that PCMP-001 Version 2.0 does not replace. Where a study departed from them, PCR-007 records the departure. Each change below names the Version 1.0 text that it replaces. Changes 1 to 10 also name the rule of PCMP-001 Version 2.0, or the action of INV-001, that they apply, and change 11 corrects descriptions of Version 1.0 that were wrong.

1. The step criteria are intermediate acceptance criteria. The Version 1.0 reports judged the pool against an aggregate limit of 1 % high molecular weight species (HMW) and a host cell protein limit of 193 ng/mg. Each limit was the drug substance criterion carried back through the nominal clearance of the later steps in this campaign and then divided by a margin, 5 for aggregate and 3.2 for host cell protein (INV-001 finding 2). The limits were set at analysis and were written into the plan text when the package was compiled in July 2026, without a revision of the approved plan (INV-001 finding 1). Version 2.0 judges the pool against the intermediate acceptance criteria of PCMP-001 §7.3, which are derived from the drug substance criteria and the platform worst cases of the downstream steps (Section 7.3). A platform worst case is the lowest clearance factor, the level entering a step divided by the level leaving it, or the largest increase that the step showed for X-Mab, Y-Mab and Z-Mab. The host cell protein criterion uses a verified A-Mab value in place of a platform worst case, the bound of 4.27 for the clearance factor of anion exchange. Table 1.1 compares the two versions. The Version 2.0 criteria for aggregate and host cell protein are wider than the Version 1.0 limits. Because the criteria rest on platform numbers, Section 4.1 no longer says that prior knowledge is not carried across as a numerical claim, or that every criterion rests on data generated under this plan.
2. Fragments are judged at this step. Version 1.0 of this plan set no criterion for fragments, and the Version 1.0 report did not judge them. Version 2.0 judges them against the drug substance criterion of 2 % (PCMP-001 §7.3), on results by non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS) under the analytical method decision AMD-001 (PCMP-001 §7.8 and Section 5.3). The decision answers INV-001 finding 6, that size exclusion chromatography

(SEC) reads fragments high in samples that still carry host cell protein. The Version 1.0 sentences that named aggregate and host cell protein as the attributes judged, or counted the responses, now include fragments. They are in Section 1, objectives 2 and 3, Section 4.2, Section 5.3, Section 7.5 and Section 8.

3. The scale-down model is qualified for each output by its offset, not by a test of equivalence. Version 1.0 of this plan read “the model will be accepted where no statistically significant difference is found on any of those attributes” and “Execution may begin once the plan is approved and the scale-down model has been accepted under §5.1”. Under PCMP-001 §7.1 the model is qualified for each output by its offset from manufacturing scale, and every judgement against a criterion is stated at manufacturing scale. The qualification runs and the offset estimates come before the first characterization run, and the decision whether the model is qualified for an output is made at analysis (Section 5.1 and Section 7.1). As PCMP-001 §7.1 requires of each step plan, this plan names the outputs compared, the number of runs at each scale and the manufacturing-scale data set. The Version 1.0 sentences that called the model qualified before the analysis, in Section 1, Section 5.1, Section 11 and Section 13, are changed to match.
4. Model adequacy is judged by the four tests of PCMP-001 §7.2. Version 1.0 of this plan judged a model on four things together. They were the overall F-test at $p < 0.05$, the test for lack of fit, the closeness of R^2, adjusted R^2 and predicted R^2, for which it set no limit, and an examination of the residuals that identified any influential run, with no test and no cut-off. Version 2.0 sets a limit of 0.20 on the difference between R^2 and predicted R^2, and states the tests on the residuals and on influential runs with their thresholds (Section 7.2). Version 1.0 of this plan read “A response whose model is not adequate will be reported as characterized by its data rather than by a model, and no design space and no proven acceptable range will be derived from it.” Under PCMP-001 §7.2 a model that fails the test for lack of fit or the limit on overfitting is either refitted with a justified change of terms or transformation, or it is not used to set a range.
5. The NOR-propagated analysis is defined. It is the second PAR analysis of Section 8, which scans one parameter while each other parameter is drawn around its set-point across its normal operating range (NOR). Version 1.0 of this plan already judged single batches in that analysis. It drew the other factors within their NORs without stating a distribution, and accepted a setting where the predicted values met the criterion across the draws. Version 2.0 draws each other factor from the normal distribution that PCMP-001 §6.4 defines, adds the uncertainty of the fitted coefficients to each draw, and accepts a setting where the 97.5th percentile of the simulated batches does not exceed the criterion.
6. The operating region is drawn on the prediction interval of a single run. Version 1.0 of this plan read “The operating region will be declared where the fitted models predict that both attributes in Table 7.1 meet their in-process criteria, evaluated over the studied ranges of the 4 multivariate factors”, which judged the mean prediction. Under PCMP-001 §6.2 a setting lies inside the operating region when every output with a criterion meets that criterion at both ends of the 95 % prediction interval of a single run (Section 7.5).
7. The platform worst cases of this step are verified. PCR-007 checks the platform worst cases of cation exchange on which the criteria upstream rely, and the verification rule of PCMP-001 §7.5 decides whether a check tightens or widens

a criterion (Section 7.6). Version 1.0 of this plan had no such check, because it had no criteria derived from platform worst cases. Objective 6, to verify those worst cases, and the verification in the deliverables of Section 11 are added for it.

8. Classification also applies two rules of PCMP-001 Version 2.0. The Version 1.0 rule, which classifies a parameter from the size of its effect and the risk that it moves outside the declared region, is kept word for word. PCMP-001 §6.2 calls an output affected by a parameter when a term that contains the parameter is significant, and PCMP-001 §7.7 asks the report to state the size of each effect beside its significance and to give the reason for each class. The change is that the words of the Version 1.0 rule, held comfortably inside the region or narrow relative to it, are judged on the effect across the PAR and on the margin of the PAR to the NOR, because neither PCMP-001 nor SOP-4001 sets a numerical threshold for them (Section 7.7).
9. Capability thresholds are set for the drug substance. As in Version 1.0, no minimum capability index is set for the outputs of this step. PCMP-001 §7.9 now sets a target and a minimum process capability index (Cpk), against which PCMR-001 judges the drug substance (Section 5.5).
10. Execution windows and calibration status are recorded. The Version 1.0 records gave no execution dates, and Version 1.0 of this plan read “No calendar dates are committed here.” The plan text compiled in July 2026 carried in its title block 2026-07-24, the date on which the Version 1.0 reports were approved (INV-001 finding 5), and it carried the step criteria that change 1 replaces. As PCMP-001 Version 2.0 (§5 and §9) and INV-001 preventive action PA3 require, this revision records the execution window of each study, that is the dates of its first and last laboratory runs, and the calibration status of the instrumented system named in its record (Section 11). Section 9 adds the term discrepancy from the plan, for work that this plan does not foresee.
11. Descriptions of Version 1.0 that were wrong are corrected, with no change to the design, and the expected directions of the four expectations of Section 4.1 are unchanged. In Section 1, Version 1.0 said that small viruses are claimed at the low-pH viral inactivation step. They are claimed at anion exchange and virus filtration only. In Section 4.1, Version 1.0 said that most host cell proteins are acidic at the operating pH. They have an isoelectric point below the load pH, and the part that binds weakly or with the bound antibody is now named. Version 1.0 said that the leached Protein A in complex with the antibody is released when the antibody binds the ligand, and that its modest clearance is a property of the chemistry. That complex binds with the antibody, which limits the clearance, and Section 4.2 now says that free leached Protein A stays unbound. Version 1.0 said that at a high load weakly bound host cell protein finds fewer free ligand sites from which the wash would displace it. At a high load more of it is held in association with the bound antibody. Version 1.0 said that yield is lost at the two edges of the pool and named two losses on the trailing edge. The loss on the leading edge is now named. In Section 4.3, Version 1.0 said that the range of elution buffer pH stops short of the pH at which the antibody would no longer bind reproducibly. It stops short of incomplete elution at its low end and of elution of aggregate with the monomer at its high end. In Section 5.1, Version 1.0 called the centre point spread the only estimate of experimental error that

the analysis uses. The residual of each fitted model is another, and the sentence is removed. In Section 5.2, Version 1.0 described the collection thresholds as relative to the peak maximum, while its Table 6.1 set the stop collect in OD units, from 0.5 to 1.5 with a set-point of 1. The description now matches the design. Version 1.0 also said that one buffer sets the ionic strength of the load and the wash. The load is the adjusted pool, and the equal conductivities are a design choice of the platform. The temperature control is named once, as controlled room temperature. In Section 6.2, Version 1.0 said that a resolution IV fraction would alias the interactions of protein load with each other. It would alias each of them with the interaction of the other two factors. In Section 12, Version 1.0 said that the qualification and the centre point performance would confirm the first assumption. The designs test it against the expectations of Section 4.1.

Table 1.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, cation exchange.

Attribute	Version 1.0[a]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Aggregates, HMW (%)	≤ 1.00	R1, margin 5.0	≤ 5.00	Drug substance
Host cell protein (ng/mg)	≤ 193	R1, margin 3.2	≤ 427	Carried back, widened[b] from ≤ 400
Fragments, LMW (%)	not assessed[c]	–	≤ 2.00	Drug substance

[a] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on a model that meets the first three adequacy criteria of PCMP-001 §7.2.
[c] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.

The derivation of these criteria uses no result of the studies of this plan (PCMP-001 §7.3).

2 Objectives

The study has six objectives.

1. To quantify the relationship between each studied parameter and each response of the step, over ranges wider than the ranges in which the step will be operated.
2. To define the multivariate region over which the fitted models predict that the aggregate, the host cell protein and the fragment content of the eluate pool all meet the criteria in Table 7.1.
3. To establish a proven acceptable range for each parameter against the aggregate, the host cell protein and the fragment content of the pool, both with the rest of the step at target and with the rest of the step varying within its normal operating ranges.
4. To classify each parameter under SOP-4001, from the size of its demonstrated effect and from the capability with which the parameter is controlled.
5. To state the ranges, in-process controls and monitoring this step contributes to the control strategy, and to justify them for Stage 2.
6. To verify the platform worst cases of cation exchange on which the criteria upstream rely (Section 7.6). This objective is added at Version 2.0.

The report PCR-007 will present the outcome against each of these objectives.

3 Related documents

Table 3.1 lists the documents this plan depends on and the document it delivers. The risk assessment RA-001 set the scope of the study, this plan executes that scope, and PCR-007 will report the outcome. The controlled procedures and method validations that govern the work are listed in Table 3.2.

Table 3.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

The SOP and AMV numbers are placeholders.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3014	Residual DNA (qPCR)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3020	Purity (non-reduced CE-SDS)

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The cation exchange step is operated in bind and elute mode and is the principal aggregate polishing operation of the purification train. Below its isoelectric point the antibody carries a net positive charge and binds the sulfonate ligand of the resin. It is eluted by raising the pH of the elution buffer, which weakens the electrostatic attraction. Aggregate presents more positive surface to the ligand than monomer does and is therefore retained more strongly, so it elutes later than monomer. The clearance the step achieves depends on how well the two species are resolved and on the point at which pool collection is stopped.

Host cell protein, DNA and leached Protein A are cleared because they carry a net negative charge at the load pH and are therefore not retained by the sulfonate ligand. Most host cell proteins have an isoelectric point below the load pH, so they leave in the load flow through, and the weakly bound remainder is displaced during the wash. A part of the host cell protein is held in association with the bound antibody, where the wash does not displace it, and it elutes with the antibody. DNA carries a strong negative charge and is repelled by the ligand, so it passes through in the load and its clearance is large and insensitive to the operating parameters. Leached Protein A and its fragments are acidic as well and do not bind under load conditions, but the leached Protein A that is in complex with the antibody binds with it, which limits the clearance. The step therefore clears leached Protein A by a modest factor, which is not expected to depend on any parameter studied here.

Yield is lost at the two edges of the pool. Monomer that elutes before collection starts is lost on the leading edge, and monomer that elutes after the stop collect point, or with the aggregate that is cut away, is lost on the trailing edge.

Fragments carry less positive charge than intact antibody at the binding pH and elute on the leading side of the main peak, so the leading fractions are rich in fragments. A later start of collection leaves more of that edge out of the pool, and a later stop adds more of the trailing eluate, which carries few fragments. Both lower the fragment content of the pool. The step is expected to clear part of the fragment load, not all of it.

The step has been operated on the Novacyte humanized immunoglobulin G1 (IgG1) platform for three previous products (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical and engineering manufacture (CMC Biotech Working Group 2009). The resin chemistry, the buffer system and the mode of operation are unchanged for A-Mab. This experience establishes the mechanisms above and identifies the parameters worth studying. Prior knowledge also supplies the platform worst cases from

which the criteria are derived (Section 7.3 and Section 7.6). Every range reported in PCR-007 will rest on data generated under this plan.

Four expectations follow from that experience, and the multivariate design is built to test them.

- A higher protein load occupies more of the sulfonate ligand, so at the top of the studied range the column approaches its dynamic binding capacity. As occupancy rises the monomer and the aggregate peak both broaden and overlap further, and a larger share of the host cell protein is held on the column through its association with the bound antibody, so that it escapes the wash and elutes with the antibody. Rising ligand occupancy is therefore expected to move aggregate clearance, host cell protein clearance and yield in the same direction, all three falling as load rises.
- Load/wash conductivity sets the ionic strength that shields the charge on the ligand and on the bound species. A higher conductivity is expected to strip more of the weakly bound host cell protein during the wash, while the antibody, which binds more strongly, stays bound. Its effect should fall on impurity clearance rather than on the aggregate separation, which happens during elution.
- Elution buffer pH sets the net charge on the antibody at the moment of elution. A higher pH within the studied range weakens binding and brings monomer and aggregate off the column earlier and closer together, so it is expected to reduce the resolution on which aggregate clearance depends.
- Monomer elutes ahead of aggregate, so the material still leaving the column on the descending edge of the peak is progressively richer in aggregate. Stopping collection later therefore recovers more monomer from the tail and carries the leading part of the aggregate peak into the pool, and stopping earlier leaves both behind.

The stop collect point is an absorbance on the descending edge (Table 6.1), so a lower setting stops collection later. At a high load more host cell protein is held in association with the bound antibody, and a higher ionic strength weakens that association, so a higher load/wash conductivity is expected to remove more host cell protein at a high load than at a low one.

Elution flow rate sets the residence time of the antibody in the pores of the resin during elution, and a longer residence time lets the antibody equilibrate more completely with the mobile phase and so sharpens the elution peak. Over the range in Table 6.1, and at the particle size of the platform resin, the change in peak sharpness is expected to be too small to move the aggregate content of the pool.

4.2 Quality attributes in scope

The drug substance register of critical quality attributes (CQAs) assigns no quality attribute to the cation exchange step. Aggregate is formed in the production bioreactor and is added to during the low-pH hold, host cell protein and DNA enter with the harvested cell culture fluid, leached Protein A is released from the capture resin,

and fragments are formed in culture and during the harvest hold. This step removes them. Table 4.1 lists the five attributes in scope with their drug substance acceptance criteria and the criticality assigned to each. Criticality follows the continuum of ICH Q9 rather than a binary split (International Council for Harmonisation 2023b), and it was assigned with Tool #1 of the case study, which combines the impact of the attribute on safety and efficacy with the uncertainty in that impact (CMC Biotech Working Group 2009).

Table 4.1: Quality attributes in scope at cation exchange, with their drug substance criteria and criticality.

Category	Quality attribute	Drug substance criterion	Criticality	Tool #1 score
Size variant	Aggregates (HMW)	≤ 5 % HMW (SEC)	H	60
	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12
Process impurity	Host Cell Protein (HCP)	≤ 100 ng/mg	M-H	36
	Residual DNA	≤ 0.001 ng/dose	VL	6
	Leached Protein A	≤ 5 ppm	L-M	16

VL, very low. L, low. M, moderate. H, high. VH, very high. A paired code such as M-H means moderate to high.

Three of the five attributes will be measured as responses of the designed experiments. Aggregate is the attribute the step exists to control and carries the highest criticality of the five. Host cell protein is cleared here and again at the anion exchange step, so this step must hand on a pool that the remaining train can bring to the drug substance criterion. Fragments are measured by nrCE-SDS in the response-surface design (Section 5.3). Residual DNA and leached Protein A will be measured across the step so that their clearance is confirmed, and neither is treated as a response of the designed experiments, because DNA is repelled by the sulfonate ligand at every conductivity and pH in Table 6.1 and free leached Protein A stays unbound over the same range. PCR-007 will report the clearance observed for both, and will claim a parameter dependence for neither unless the data show one.

Step yield is a process performance attribute rather than a quality attribute. It is measured on every run because the monomer that is cut away with the aggregate is lost from the pool, so the aggregate clearance and the yield of a setting have to be read together. It has no acceptance criterion in Table 4.1.

4.3 Risk-based prioritization of parameters

All process parameters of the cation exchange step were carried into the risk ranking and filtering exercise of RA-001, in which each parameter was given one ranking for its potential impact on a quality attribute and a second for its potential to interact with other parameters, and the two rankings were combined into a study assignment

(CMC Biotech Working Group 2009). Where no data or rationale were available to make an assessment, the parameter was ranked at the highest level.

The assignment produced three groups. Four parameters were assigned to the multivariate design. Protein load, elution buffer pH and elution stop collect each move the aggregate peak relative to the point at which collection stops, and protein load acts with the load/wash conductivity to set how much weakly bound host cell protein survives the wash, so the effects of the four on the pool cannot be assumed to add. Elution flow rate was assigned to a univariate assessment, since its expected effect on the separation is small (Section 4.1). Parameters that are fixed by equipment design or held to platform specification, such as bed height, column packing quality and buffer composition, were not assigned a new study. Their ranges are supported by the platform knowledge space and by the operation and life cycle procedures in Table 3.2.

The ranges in Table 6.1 are wider than the normal operating ranges in every case. Each was set from the control capability of the commercial equipment and from the range over which the platform has been operated, so that the study bounds the step beyond the region in which it will run and supports later movement within the region it defines (International Council for Harmonisation 2009). Ranges were not widened past the point at which the step would fail for a reason that is already understood. The elution buffer pH range stops short of the pH at which the antibody would no longer elute completely, at its low end, or would elute together with its aggregate, at its high end, because a run at such a pH would confirm the mechanism and would tell the study nothing about the operating region.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run on a laboratory column packed with the same resin type and to the same bed height as the commercial column, and operated at the same linear velocities. Protein load will be set in grams of antibody per litre of resin, and the equilibration, wash, elution and strip volumes will be set in column volumes, so that every parameter of the study is expressed at its commercial equivalent. Packing quality will be verified before each campaign by plate count and peak asymmetry, and a column that fails either test will be repacked before use. The model and its operation are described in SOP-1001 and SOP-2010.

The qualification runs will be made against commercial scale data before any characterization run is executed, and the offset of each output will be estimated from them. Qualification will compare the model and the commercial step at the set-point condition of Table 6.1 on the attributes that enter and leave the step, which are step yield, pool aggregate, pool host cell protein, residual DNA and leached Protein A. The comparison will be made from replicate runs at small scale against the available at-scale batches, 3 scale-down runs at set-point against 5 A-Mab batches at manufacturing scale. Plate count and peak asymmetry are compared as well (PCMP-001 §5).

The offset of an output is the mean of the scale-down runs at set-point minus the mean of the manufacturing-scale batches, with its two-sided 90 % confidence interval. Each end of that interval is a one-sided 95 % confidence bound. Whether the model is qualified for an output is decided at analysis, by the rule of Section 7.1, because that decision needs the response-surface model. Pool aggregate and pool host cell protein have a criterion at this step, and every judgement of them against it first subtracts the offset, so their ranges and conclusions are stated at manufacturing scale. The other outputs compared have no criterion at this step, so their offsets are recorded and no conclusion depends on them.

The qualification does not compare fragments at both scales. Their offset is unknown, not zero, and fragments are judged as the scale-down model gives them (PCMP-001 §7.1). That judgement rests on the assumption, stated in Section 12, that fragments separate in the same way at both scales.

Reproducibility of the model will be re-established inside each designed experiment. The centre point runs of both designs are executed at the set-point of every factor, so their spread measures the reproducibility of the model under study conditions and provides the pure error term against which lack of fit is tested (see Section 5.5).

The claims this study makes are bounded by the scale-down model. A scale-down column reproduces the chromatography of the commercial column, and it does not reproduce the hold times, the pool volumes or the mixing of the commercial suite. Conclusions drawn here apply to the commercial step through the qualification above, and they will be confirmed at scale during Stage 2.

5.2 Operation

The step receives the neutralized pool of the low-pH viral inactivation step. The pool is adjusted to the load conductivity given in Table 6.1 and is loaded onto the equilibrated column at the target protein load. The column is then washed at the same conductivity as the load. The platform sets the two equal by design, so they are varied together and cannot be varied independently. Elution is a step change to the elution buffer at the pH under study. The pool is collected between a threshold on the ascending edge of the elution peak and the stop collect threshold on the descending edge, both read as ultraviolet absorbance (OD280). After collection the column is stripped, sanitized and re-equilibrated under SOP-2008.

Buffer preparation, column packing and resin storage are held to platform specification and are governed by the procedures in Table 3.2. They are identity and quality controlled inputs to this study rather than factors of it. Chromatography is performed at controlled room temperature and temperature is not varied, because the platform operates the step under the same control at both sites and the transfer introduces no difference in that control.

5.3 Analytical methods

Every response and every attribute in scope will be measured by a validated method. Table 3.2 lists the method validation reports that apply. Pool aggregate will be measured by size exclusion chromatography under AMV-3011 and reported as the percentage of high molecular weight species. Host cell protein will be measured by enzyme-linked immunosorbent assay under AMV-3012 and reported per milligram of antibody, so that the reported value does not move with the concentration of the pool. Residual DNA will be measured by quantitative polymerase chain reaction under AMV-3014, and leached Protein A by enzyme-linked immunosorbent assay under AMV-3016. Step yield will be calculated from the antibody mass in the pool and in the load, each determined by ultraviolet absorbance against the extinction coefficient of A-Mab.

Fragments are measured by nrCE-SDS under AMV-3020. From 2026-03-20, every study that measures fragments, at every step, measures them by nrCE-SDS, following the analytical method decision AMD-001, so that one method spans the train (PCMP-001 §7.8). PCMP-001 Table 7.3 shows which method each design of the campaign used. Fragments are judged on the nrCE-SDS results only.

Assay variability is part of the residual variance of each fitted model, and it is part of the centre point spread that estimates pure error, so a response measured by a less precise assay will carry a wider prediction interval and a less sensitive lack of fit test. Where a predicted value sits within assay variability of its criterion, PCR-007 should attribute the result to the assay and should not claim an effect. Host cell protein is expected to be the least reproducible of the responses, and the centre point runs will establish whether that expectation holds.

5.4 Sampling plan

Each run will be sampled at four points. The load will be sampled after conditioning, so that the aggregate, host cell protein, DNA and leached Protein A levels entering the step are known for every run. The flow through and the wash will each be sampled as a single pooled fraction, which allows the host cell protein mass balance across the step to be closed. The eluate pool will be sampled after collection and mixing are complete, and it carries every response of the study. A strip fraction will be sampled and retained, and will be tested only where a run fails to close its mass balance.

Replication is concentrated at the centre point. The screening design carries 3 centre point runs and the response-surface design carries 4. Samples will be assigned to analytical runs in an order that does not follow the design order, so that drift in an assay cannot align with a factor. Retained samples will be held under the conditions and for the period stated in SOP-2010, so that a re-test is possible where a result is questioned.

5.5 Statistical methods

All factors will be analysed in coded units. Each factor is coded so that its set-point is 0 and the edges of its studied range are -1 and $+1$, which makes the estimated effects comparable across factors carrying different units and makes the intercept the predicted response at the centre of the design. The letter codes are given in Table 6.2 and are used for the terms of both models.

The screening design will be analysed by ordinary least squares with all main effects and all two-factor interactions. An effect will be reported as twice the coded coefficient, which is the change in the response across the full studied range of that factor. Terms will be judged significant at $p < 0.05$. The purpose of the screening analysis is to identify which factors and which interactions change each response, and by how much across the studied range. It is not the predictive model of the step, since a model without quadratic terms cannot describe curvature, and the response-surface model is the model from which the operating region, the proven acceptable ranges and the capability estimate will all be derived. The judgement of each output over the NOR (Section 7.4) and the verification of the platform worst cases (Section 7.6) also come from it.

The response-surface design will be analysed as a full quadratic model in its 4 factors, with main effects, two-factor interactions and pure quadratic terms, 15 terms with the intercept. Each fitted model is reported with R^2, adjusted R^2 and predicted R^2, the last computed from the prediction error sum of squares (PRESS), with the model F-test and with the test for lack of fit against the pure error of the centre point runs. Model adequacy is judged by the four tests of Section 7.2.

A factor with no significant term in either model will be reported as having no demonstrated effect on that response. Such a statement is bounded by the range studied and by the power the design carries, and PCR-007 will state it in those terms rather than as an absence of effect in general.

The 95 % prediction interval of a single run that Section 7.4 uses is exact. It uses the t distribution and includes the residual variance and the uncertainty of the fitted coefficients. For pool aggregate and pool host cell protein it is computed at manufacturing scale, by subtracting the offset of Section 5.1.

Capability at commercial scale will be estimated by Monte Carlo simulation. Two thousand batches will be simulated with every parameter drawn within its normal operating range, the fitted models will be evaluated at each draw, and a one-sided capability index will be computed for each attribute in Table 7.1 against its criterion. No minimum capability index is set as an acceptance criterion in this plan. The decision rule is the one stated in Section 7, and the capability estimate will be reported alongside it, with the margin between the predicted distribution and the limit, as supporting evidence. The capability of the drug substance is estimated from the integrated model of the train that PCMP-001 §6.5 defines. PCMR-001 judges it against the campaign thresholds of PCMP-001 §7.9, a target process capability index (Cpk) of 1.33 and a minimum of 1.0.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters of the step with their set-points, the range each will be studied over, the normal operating range and the study assigned to it.

Table 6.1: Parameters of the cation exchange step, with the range studied, the NOR and the study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10–40	18–32	Multivariate
Load/Wash conductivity	mS/cm	5	3–7	4–6	Multivariate
Elution buffer pH	–	6.00	5.8–6.2	5.85–6.15	Multivariate
Elution stop collect	AU (OD280, descending)	1.0	0.5–1.5	0.7–1.3	Multivariate
Elution flow rate	cm/hr	200	100–300	150–250	Univariate

The 4 multivariate parameters enter both designs as the factors of Table 6.2. The letter codes are used for the model terms in PCR-007 and for the design matrices in Appendix A and Appendix B.

Table 6.2: Coded factors of the screening and response-surface designs.

Code	Factor	Unit
A	Protein load	g/L resin
B	Load/Wash conductivity	mS/cm
C	Elution buffer pH	–
D	Elution stop collect	AU (OD280, descending)

6.2 Screening design

The screening study is a two-level full factorial in the 4 multivariate factors, augmented with 3 centre point runs, giving 19 runs in total. A full factorial was selected rather than a fraction, because the factor count is small and because protein load is expected to interact with each of the other three factors, through the peak broadening that rising ligand occupancy produces. A resolution IV fraction would alias each

of those interactions with the interaction of the other two factors (AB with CD, AC with BD and AD with BC) and would leave the analysis unable to say which of them is active. The design as planned estimates every main effect and every two-factor interaction independently.

Runs will be executed in a randomized order, with the centre point runs distributed through the campaign rather than grouped, so that drift in the model or in the feed material appears in the centre point spread rather than in a factor effect. The planned coded design is given in Appendix A. All runs will be performed on a single conditioned feed lot where the available material allows it. Where more than one lot is required, the lot will be recorded with the run and the centre point runs will be balanced across lots.

6.3 Response-surface design

The response-surface study is a face-centred central composite design in the 4 factors of the screening study, with 16 factorial runs, 8 axial runs and 4 centre point runs, giving 28 runs in total. The axial runs are placed on the faces of the cube rather than beyond them, so that no run is executed outside the studied range of any factor. The studied ranges are already wider than the normal operating ranges, and a rotatable design would require settings outside them for which the step has no prior support.

The response-surface design adds the quadratic terms that the screening design cannot estimate. Curvature is expected on elution stop collect and on elution buffer pH. Both change where the pool is cut relative to the aggregate peak, and because that peak has a maximum, equal steps in the cut point add unequal amounts of aggregate to the pool. The planned coded design is given in Appendix B.

The two designs are executed in sequence and are analysed separately. Should the screening analysis identify an active factor that is not carried into the response-surface design, the design should be augmented before the operating region is declared, and the augmentation should be documented as a change to this plan under Section 13. The response-surface design carries all 4 screening factors, so this rule cannot apply at this step.

6.4 Univariate assessment

Elution flow rate will be assessed one factor at a time. Runs will be executed at the low edge, the set-point and the high edge of the range given in Table 6.1, with every other parameter held at its set-point, and the responses that Version 1.0 listed in its §5.3 (pool aggregate, pool host cell protein, residual DNA, leached Protein A and step yield) will be measured on each. The assessment will establish whether a longer residence time sharpens the elution peak enough to change the aggregate content or the yield of the pool over the studied range, and it will support the range within which flow rate is controlled in manufacture.

A univariate assessment cannot determine whether flow rate interacts with the other parameters. It is accepted on that basis here because flow rate changes the residence time of the antibody in the pores of the resin and with it the sharpness of the elution peak, and it changes neither the net charge on the antibody nor the ionic strength that sets how strongly the antibody is held, so an interaction with load, conductivity, elution pH or the collection window is not expected. Should the assessment show an effect on any response comparable in size with the effects estimated in the screening study, flow rate should be carried into a multivariate study before the operating region is declared.

7 Acceptance and decision criteria

Version 1.0 of this plan named four decisions, each with its own rule. They are the acceptance of the scale-down model, the adequacy of a model, the operating region and the classification of each parameter. PCMP-001 Version 2.0 replaces the first, extends the second and the third, and adds the judgement of each output against its criterion and the verification of the platform worst cases. The proven acceptable ranges follow in Section 8.

Several decisions of Version 1.0 rest on words for which it set no numerical threshold. They are the claim of a parameter dependence for residual DNA and leached Protein A “unless the data show one” (Section 4.2), the repacking of a column that fails the packing tests (Section 5.1), the attribution of a result “within assay variability of its criterion” to the assay (Section 5.3), the test of the strip fraction when a run fails to close its mass balance (Section 5.4), the move of elution flow rate into a multivariate study after an effect “comparable in size with the effects estimated in the screening study” (Section 6.4) and the identification of the runs made on a lot “out of trend” (Section 12). This revision sets none either, and PCR-007 states the approach it takes for each.

7.1 Qualification of the scale-down model

The scale-down model is qualified for each output separately, by its offset from manufacturing scale and not by a test of equivalence (PCMP-001 §7.1). For pool aggregate and pool host cell protein, the outputs compared at both scales that have a criterion at this step, the model is qualified when the conclusion over the NOR is the same with the offset at either end of its 90 % confidence interval. The conclusion over the NOR is whether the output meets its criterion at every combination of NOR low, set-point and NOR high of the 4 factors, judged as Section 7.4 states. A model whose conclusion changes between the two ends of the interval is not used for that output. Where the cause of an offset is known, PCR-007 states it. Where the model is not qualified for an output, PCR-007 says so and states the consequence.

7.2 Model adequacy

PCMP-001 §7.2 sets four criteria for the adequacy of a response-surface model, and this plan states them as tests.

1. The model F-test is significant at α = 0.05, the level PCMP-001 sets for model terms.
2. Lack of fit against the pure error of the centre point runs is not significant at α = 0.05.
3. Predicted R^2 falls short of R^2 by no more than 0.20, the limit on overfitting that PCMP-001 sets. A model that fits its own runs much better than it predicts runs left out of the fit has more terms than the data support.
4. The residuals show no pattern against the predicted values or the run order, and no single run dominates the fit. The absolute residuals are regressed on the predicted values, and the residuals on the run order, and neither slope may be significant at α = 0.05. No run may have a Cook's distance above 4/n, where n is the number of runs in the fit.

PCMP-001 §7.2 states the fourth criterion without a statistic. The two tests and their thresholds are those of the campaign, applied in the same way in every step plan. The absolute residuals are regressed because the residuals of a least squares fit have no slope against the predicted values by construction, while a scatter that grows with the level shows as a slope of their absolute values.

A model that fails the test for lack of fit or the limit on overfitting is either refitted with a justified change of terms or transformation, or it is not used to set a range (PCMP-001 §7.2). PCR-007 examines any run that the fourth test flags, and Section 7.6 states which of the tests the model behind a widening bound must pass. Where no adequate model results, PCR-007 should report the response descriptively, against the criterion, over the runs of the design. Where lack of fit is significant, PCR-007 should state whether the cause is a missing term or an outlying run, and should establish that from the residuals rather than from the summary statistics alone.

7.3 Intermediate acceptance criteria

The quality criteria the pool is judged against are given in Table 7.1. Each is an intermediate acceptance criterion (IAC). PCMP-001 §7.3 derives it by carrying the drug substance criterion backward through the train, step by step, with the platform worst case of each downstream step, and no margin is applied.

No step after cation exchange changes aggregate or fragments in this derivation. The cation exchange pool is therefore judged against the drug substance criteria themselves, 5 % for aggregate and 2 % for fragments. For host cell protein the drug substance criterion of 100 ng/mg is carried back through anion exchange. With the platform worst case of anion exchange, a clearance factor of 4, the criterion is 400 ng/mg a priori. The anion exchange report PCR-008 verified that worst case. It found a one-sided 95 % prediction bound of a single batch of 4.27 for the clearance factor at the least favourable combination of the anion exchange NOR, better than the platform value, on a model that meets the first three adequacy criteria of PCMP-001 §7.2 and is not an exact fit. The bound therefore replaced the platform value, and the criterion in force is 427 ng/mg. PCR-008, the report that gave the bound, examines it and argues the change by its risk (PCMP-001 §7.5).

Table 7.1: Intermediate acceptance criteria for the cation exchange output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Host cell protein (ng/mg)	≤ 100	4.27	0.00	≤ 427	Carried back, widened[a)]
Aggregates, HMW (%)	≤ 5.00	1.00	0.00	≤ 5.00	DS criterion[b)]
Fragments, LMW (%)	≤ 2.00	1.00	0.00	≤ 2.00	DS criterion[b)]

[a)] The one-sided 95 % prediction bound of a single batch at a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.
[b)] No downstream step changes the attribute, so the drug-substance criterion applies.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

Residual DNA, leached Protein A and step yield have no criterion at this step, so no range is derived from them. The platform history gives no worst-case DNA clearance for the steps after Protein A, so no DNA criterion can be carried back, and residual DNA is judged on the drug substance only. Leached Protein A is judged in the Protein A pool, against a criterion carried back through the platform worst cases of cation exchange and anion exchange.

7.4 Judgement of an output against its criterion

Every output with a criterion at this step is a quality attribute. Its criterion applies to each batch, so each output is judged on a single batch. A response-surface model predicts the mean response, and a single batch scatters around that mean, so the judgement uses a prediction interval or bound of a single batch (Section 5.5). Four analyses judge an output in this way. They are the judgement over the NOR, the operating region (Section 7.5), the NOR-propagated PAR (Section 8) and the verification of the platform worst cases (Section 7.6).

The judgement over the NOR asks whether the exact 95 % prediction interval of a single run, at manufacturing scale, lies within the criterion at every one of the 81 combinations of NOR low, set-point and NOR high of the 4 factors. The qualification of Section 7.1 uses it. All criteria at this step are upper limits, so the upper end of the interval decides.

7.5 Operating region

The operating region is the region of the 4 multivariate factors, within their studied ranges, over which every output with a criterion in Table 7.1 meets that criterion at both ends of the 95 % prediction interval of a single run, at manufacturing scale (PCMP-001 §6.2 and Section 7.4). Version 1.0 also calls it the declared region, and PCMP-001 calls it the design space of the step. The declared region will lie inside the ranges studied, and the normal operating ranges will lie inside the declared region. PCR-007 states whether they do. PCR-007 will state the margin between the two and will identify the attribute and the parameter that bound the region most tightly. Where the models predict that a criterion is met across the whole studied range of a parameter, that will be reported as the finding, and the range will not be narrowed to manufacture a margin the data do not require.

Step yield has no quality criterion and does not bound the operating region. It will be reported as a process performance attribute across the region, and a setting that meets every quality criterion at a cost in yield will be described as such rather than excluded.

7.6 Verification of the platform worst cases

The criteria upstream of this step assume that cation exchange clears each attribute at least as well as its platform worst case, the lowest clearance factor seen for X-Mab, Y-Mab and Z-Mab. Table 7.2 lists the four platform worst cases of this step and the criteria upstream that rely on each. PCR-007 checks those of host cell protein, aggregate and fragments, which are responses of the designs (PCMP-001 §7.5). Leached Protein A is not a response of the designs, so its platform value stands.

Table 7.2: Platform worst cases of cation exchange that the criteria upstream rely on.

Attribute	Platform worst case (lowest clearance factor)	Criteria upstream that rely on it	Response of the designs
Host cell protein	40	Protein A	Yes
Aggregates, HMW	2	Production bioreactor and low-pH viral inactivation	Yes
Fragments, LMW	1.3	Protein A and low-pH viral inactivation	Yes
Leached Protein A	3.16	Protein A	No

The check evaluates the response-surface model of this step at every combination of NOR low, set-point and NOR high of its 4 factors, which are the NOR and the set-points the studies were planned against. For the check, the clearance factor is the level that the nominal batch of the process model brings to this step, divided by the pool level that the response-surface model predicts. The model gives the worst clearance factor

that it predicts over these combinations, and the one-sided 95 % prediction bound of a single batch at the least favourable combination, on its unfavourable side. The assumption that cation exchange clears an attribute at least as well as its platform worst case is confirmed when that worst predicted clearance factor is no lower than the platform worst case. The values observed at the edges of the ranges studied, against the same nominal feed, are reported beside it for information. The check of a platform worst case changes a criterion upstream by the verification rule of PCMP-001 §7.5.

1. Where the worst clearance factor predicted within the NOR is lower than the platform worst case, the prediction bound of a single batch replaces the platform value. Every criterion upstream for that attribute is recalculated with it and becomes stricter.
2. Where the prediction bound of a single batch is better than the platform value, and the response-surface model behind it meets the first three adequacy criteria of Section 7.2 (a significant model, no significant lack of fit and the limit on overfitting) and is not an exact fit, the bound replaces the platform value. The fourth criterion, on the residuals and on influential runs, does not decide the change. PCR-007 examines the bound and states whether it depends on a single run. Every criterion upstream for that attribute is recalculated with it and becomes wider.
3. Otherwise the platform value stands.

The point prediction never replaces a platform value, and neither does a confidence bound of the mean, because each worst case concerns a quality attribute, which is judged on a single batch, and the platform value it may replace was itself the result of a single run.

A report whose check changes a criterion argues the change by its risk (PCMP-001 §7.5). It states who bears the risk, the patient or the producer. It states whether an excursion of the step output beyond the changed criterion would be detected, and where, by an in-process test or by a release test. It states what the downstream steps must then clear, and it gives the probability that the drug substance fails its specification in the integrated model of the train.

7.7 Parameter classification

A parameter will be classified under SOP-4001 from two things together, the size of the effect the study demonstrates on a quality attribute and the risk that the parameter moves outside the declared region in manufacture (CMC Biotech Working Group 2009). A parameter linked to a quality attribute and held comfortably inside the region by the commercial control system is a well-controlled critical process parameter. A parameter linked to a quality attribute whose control is narrow relative to the region is a critical process parameter. A parameter that changes yield or another performance attribute without changing a quality attribute is a key process parameter. A parameter that changes neither over the range studied is a general

process parameter. The classifications themselves are outcomes and are reported in PCR-007.

PCMP-001 Version 2.0 adds two rules to this one. A parameter is linked to an output, or affects it, when a term that contains the parameter is significant at α = 0.05 (PCMP-001 §6.2). PCR-007 states the size of each effect across the PAR beside its significance, and the margin of the PAR to the NOR, and gives the reason for each class on those values (PCMP-001 §7.7). Neither PCMP-001 nor SOP-4001 sets a numerical threshold for “held comfortably inside” or “narrow”, so these two values are what the words are judged on. For elution flow rate the range is the one that its univariate assessment supports (Section 8).

8 Proven acceptable ranges (planned analysis)

A proven acceptable range will be determined for each parameter of the multivariate design against the aggregate, the host cell protein and the fragment content of the pool. The criterion applied is the intermediate acceptance criterion for that attribute in Table 7.1, so a proven acceptable range in this study is the range of a parameter over which the fitted model predicts that the eluate pool meets the level the step must hand on. Elution flow rate is not a parameter of the multivariate design. Its range is supported by its own data, from the univariate assessment of Section 6.4, as PCMP-001 §4 provides for a parameter studied one factor at a time. Version 1.0 set no rule for judging that range against the criteria. The flow-rate study measures no fragments, so its data do not support the range for fragments. PCR-007 states the approach it takes for both.

Two analyses will be run for every pair of parameter and attribute, and both will be reported. The first evaluates the response-surface model across the studied range of the parameter with the other three factors held at the centre of the design, which for every parameter of this step is its set-point (Table 6.1). It answers what one parameter can do while the rest of the step is at target. The second propagates the normal operating ranges of the other three factors. At each of 81 settings across the studied range of the parameter, 2,000 draws will be taken with the other factors distributed within their normal operating ranges, and the model will be evaluated at every draw. The second analysis is the one that bounds routine operation. It will in general return the narrower of the two ranges, because a setting that is acceptable with the rest of the step at target need not be acceptable when the rest of the step sits anywhere inside its normal operating ranges.

PCMP-001 Version 2.0 calls the first the set-point analysis and the second the NOR-propagated analysis (PCMP-001 §6.4). The set-point analysis judges the mean prediction at manufacturing scale. In the NOR-propagated analysis each other factor is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, and clipped to the studied range. This assumes that the NOR spans ±3 standard deviations of the routine variation of the parameter. Each draw also takes its own set of fitted coefficients from their estimated sampling distribution, and its own residual, so the interval includes the model residual and the uncertainty of the coefficients. The 95 % prediction interval of a single batch runs from the 2.5th to the 97.5th percentile of the simulated batches, and a setting is accepted where the 97.5th percentile, at manufacturing scale, does not exceed the criterion. The NOR-propagated PAR is the one that supports a claim.

The reported proven acceptable range is the contiguous interval containing the set-point over which the criterion is met. Where the acceptable settings are not contiguous, the contiguous part containing the set-point will be reported and PCR-007 will state that this has been done. Where the criterion is met across the whole studied range, the proven acceptable range is the studied range, and the claim is bounded by that range and no wider. A parameter that affects several attributes has a PAR for each, and the range carried into the control strategy is their intersection. Where an output does not meet its criterion even at the set-point, PCR-007 reports that no PAR exists for that output and states the consequence for the NOR (PCMP-001 §7.6).

Each output with a criterion in Table 7.1 will be reported with a plot of the predicted response against the parameter carrying the largest main effect on it, with the acceptable region shaded, the set-point marked and the normal operating range marked. The Monte Carlo component uses the fixed seed 20240724, so a repeat of the analysis on the recorded data returns the same ranges.

A proven acceptable range is a statement about one parameter while the others are held at defined settings. The operating region is a statement about the 4 factors together, and a combination of settings that are each inside their proven acceptable range may still fall outside the region where an interaction is active. Both will be reported, and the control strategy will be built on the region.

9 Data management and integrity

Data generated under this plan will be recorded and retained under the ALCOA+ principles, so that every result is attributable, legible, contemporaneous, original and accurate, and so that the record is complete, consistent, enduring and available. Run conditions and observations will be recorded in the electronic laboratory notebook at the time of execution. Chromatograms and their integration will be retained in the chromatography data system under audit trail. Analytical results will be transferred from the validated instrument systems into the study dataset without manual transcription wherever the systems support it, and any manual entry will be verified by a second person.

The analysis dataset, the analysis scripts and the fitted models will be retained with the report, so that every number in PCR-007 can be traced to the run that produced it. A recorded value may be changed after review only under the correction procedure, with the original value, the reason and the author retained. Deviations from this plan will be recorded as they occur and will be reported in PCR-007 with their investigation, their root cause and their impact on the conclusions, whether or not a conclusion changes as a result.

Work that this plan does not foresee, and that no revision of it covers, is a discrepancy from the plan and not a deviation. PCR-007 records each discrepancy with what the plan foresaw, what was done and why. The date of every laboratory run is recorded, so that the execution window of each study is part of the record (Section 11).

10 Roles and responsibilities

Process Development and MSAT own this plan. They design the studies, execute the runs on the scale-down model, analyse the data and author PCR-007. The other functions in Table 10.1 review and approve within their own competence, and no function approves its own work.

Table 10.1: Functions and their responsibilities under this plan.

Function	Responsibility
Process Development / MSAT, Cambridge, MA	Owns the plan and its report, PCR-007
Analytical Development	Runs the validated methods, releases results, advises on assay variability and owns the fragment method
Manufacturing Science, Grafton, WI	Owns the qualification of the scale-down model against manufacturing scale, and confirms that the proposed ranges are ranges the commercial equipment can hold
Statistics	Reviews the designs before execution and the fitted models after it
Quality Assurance	Approves the plan and the report, reviews deviations raised under the plan

A role may be held by more than one person, and one person may hold more than one role, provided that the person who reviews an analysis is not the person who performed it. The names of the individuals holding each role at execution are recorded in the study file rather than in this plan, so that a change of personnel does not require a revision of the plan.

11 Deliverables and schedule

The deliverable of this plan is PCR-007. It will report the execution of every study described here, the models fitted to the data, the operating region, the proven acceptable ranges, the capability estimate and the classification of each parameter. It will also carry the design matrices as executed, including any run that was repeated, and every deviation recorded during the campaign. PCR-007 rolls up into PCMR-001, which consolidates the characterization of the whole train. Under Version 2.0, PCR-007 also reports the offsets of the scale-down model and its qualification for each output, the verification of the platform worst cases of this step and every discrepancy from this plan.

The work runs in five phases. The qualification runs of the scale-down model are made first, and no characterization run is executed until they are complete and the offsets estimated. The screening study follows, and its analysis identifies the factors and interactions that change each response. The response-surface study follows the screening analysis. The univariate assessment of elution flow rate may run in parallel with either designed study, since it shares no material or equipment constraint with them. Analysis, the proven acceptable range determination and the capability estimate are performed on the complete dataset, and the report is authored last.

The schedule is held in PTP-001, which sequences this study against the other characterization studies of the campaign and against the Stage 2 activities that depend on them.

Table 11.1 records the execution window of each study. The univariate assessment ran in parallel with the response-surface study.

Table 11.1: Execution windows of the studies under this plan.

Activity	Study	Start	End
Qualification runs and offset estimate	S-CEX-SDM	2026-02-04	2026-02-13
Screening design	S-CEX-SCR	2026-02-16	2026-02-27
Response-surface design	S-CEX-RSM	2026-03-02	2026-03-27
Univariate assessment of elution flow rate	S-CEX-UNI	2026-03-16	2026-03-20

The record names an instrumented system for the response-surface study only, the temperature-controlled chromatography chamber EQ-CHR-118, one of the systems that PCMP-001 §5 lists under calibration and change control. The chamber was in calibration for the whole study, which closed on 2026-03-27, before its next calibration fell due on 2026-05-30.

12 Risks and assumptions

Four risks to the study are recognized, and each carries a response.

The scale-down model may fail to represent the commercial column in a respect the qualification does not test. The qualification of Section 5.1 compares the attributes that enter and leave the step. It does not compare the flow distribution inside the two columns, which plate count and peak asymmetry bound only indirectly. Should a difference in packing quality or flow distribution be identified later, the affected conclusions would have to be revisited at scale.

The feed material may not be representative. The load for these studies is the neutralized pool of the low-pH viral inactivation step, and its aggregate and host cell protein content set the starting point of every clearance the study measures. Load attributes will be measured on every run and reported with the results, so that a clearance can be distinguished from a low starting level. Where a lot proves out of trend on any load attribute, the runs performed on it should be identified in PCR-007 and their influence on the fitted models assessed.

The resin may behave differently at a different point in its life cycle. All characterization runs will be performed on resin within the cycle count qualified under SOP-2008, and the conclusions of this study are bounded by that cycle count. Performance at the end of the qualified life is established by the life cycle study and not here.

Assay variability may obscure a small effect. The centre point replicates give the reproducibility of each response under study conditions, and an effect smaller than that spread will not be claimed.

Three assumptions are made. The first is that the platform mechanisms apply to A-Mab at this step, which the designs will test against the expectations of Section 4.1 rather than assume. The second is that the parameters in Table 6.1 are the parameters that matter, which the risk assessment RA-001 established and which this study can test only within its own factor set. The third is that a quadratic model describes the responses adequately over the studied ranges, which the lack of fit test of Section 7.2 tests directly.

Version 2.0 adds two assumptions. The fourth is that the NOR of each parameter spans ±3 standard deviations of its routine variation, on which the NOR-propagated analysis rests (Section 8). The fifth is that fragments separate in the same way at both scales, so that their judgement without an offset holds at manufacturing scale (Section 5.1). The scale-down column keeps the bed height, the linear velocity and the collection thresholds of the commercial column, and its packing passes the same tests of plate count and peak asymmetry.

13 Approvals

This plan is approved by the functions recorded in the approval record at the front of this document. Execution may begin once the plan is approved and the qualification runs of Section 5.1 have been made and the offsets estimated. Any change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan carrying the same approvals, and the revision should be approved before the affected runs are executed. This Version 2.0 is the amendment that INV-001 required as its corrective action CA2. It was issued after the runs and approved on its effective date, 2026-09-30.

Appendix A. Planned screening design matrix

Table A.1 gives the planned screening design, in the coded levels and the letter codes of Table 6.2, with −1 and +1 at the edges of the studied range of each factor. The run numbers are labels rather than an execution sequence, since execution order will be randomized.

Table A.1: Planned screening design matrix in coded levels.

Run type	Run	A	B	C	D
Factorial	1	−1	−1	−1	−1
	2	−1	−1	−1	+1
	3	−1	−1	+1	−1
	4	−1	−1	+1	+1
	5	−1	+1	−1	−1
	6	−1	+1	−1	+1
	7	−1	+1	+1	−1
	8	−1	+1	+1	+1
	9	+1	−1	−1	−1
	10	+1	−1	−1	+1
	11	+1	−1	+1	−1
	12	+1	−1	+1	+1
	13	+1	+1	−1	−1
	14	+1	+1	−1	+1
	15	+1	+1	+1	−1
	16	+1	+1	+1	+1
Centre	17	0	0	0	0
	18	0	0	0	0
	19	0	0	0	0

Appendix B. Planned response-surface design matrix

Table B.1 gives the planned response-surface design, in the same coded levels and letter codes. The factorial runs repeat the corners of the screening design, and the axial runs sit on the faces of the cube at the edge of each factor's studied range.

Table B.1: Planned response-surface design matrix in coded levels.

Run type	Run	A	B	C	D
Factorial	1	−1	−1	−1	−1
	2	−1	−1	−1	+1
	3	−1	−1	+1	−1
	4	−1	−1	+1	+1
	5	−1	+1	−1	−1
	6	−1	+1	−1	+1
	7	−1	+1	+1	−1
	8	−1	+1	+1	+1
	9	+1	−1	−1	−1
	10	+1	−1	−1	+1
	11	+1	−1	+1	−1
	12	+1	−1	+1	+1
	13	+1	+1	−1	−1
	14	+1	+1	−1	+1
	15	+1	+1	+1	−1
	16	+1	+1	+1	+1
Axial	17	−1	0	0	0
	18	+1	0	0	0
	19	0	−1	0	0
	20	0	+1	0	0
	21	0	0	−1	0
	22	0	0	+1	0
	23	0	0	0	−1
	24	0	0	0	+1
Centre	25	0	0	0	0
	26	0	0	0	0
	27	0	0	0	0
	28	0	0	0	0

References

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